

2023

Time : 3 hours

Full Marks : 60

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer any five questions,

in which question no.1 is compulsory.

1. Answer any three questions of the following :

4×3=12

- (a) Define the inner product in a vector space.
- (b) What are the properties of eigenvalues of unitary operators?
- (c) Differentiate real and complex numbers.

- (d) What is Minkowski space?
 - (e) Brief Similarity transformation.
 - (f) Explain Hamilton's principle.
2. Using the Euler-Lagrange equation, derive the equation of motion for a simple pendulum. What assumptions are made in this derivation. 12
3. Deduce the Lorentz transformations due to rotation of axes in the four dimensional Minkowski space. 12
4. Describe the Hermitian operators and their key properties. Include examples of Hermitian operators and their matrix representations. 12
5. Define eigen values and eigen vectors of a linear operator. Provide an example to explain the process of finding eigen values and eigenvectors. 12
6. What are symmetric and antisymmetric tensors? Describe how tensor components

transform under a change of co-ordinate system.

7. Define linear independence in the context of vector spaces. Provide an example to illustrate how to determine if a set of vectors is linearly independent or not. 12

8. Explain the idea of a vector space over the fields of real numbers and complex numbers with examples, and discuss their similarities and differences. 12

9. Write short notes on any two of the following :

6×2=12

(a) Euler Lagrange equation of motion

(b) Tensors

(c) Singular and non-singular transformation

(d) Isomorphisms

_____ x _____